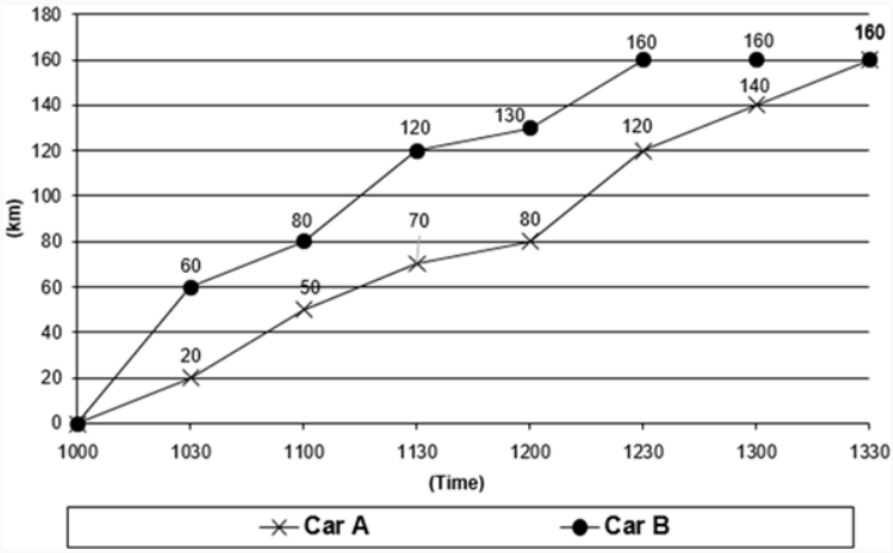
Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

The following graph provides the details about the distance covered by two vehicles travelling in the same direction, from 1000 hrs to 1330 hrs.



Directions for questions 1 and 2: Type in your answer in the input box provided below the question.

What is the average speed (in kmph) of car A in the first two hours?

Your Answer: Unattempted

Show Correct Answer | Hide Solution | PREVIOUS | NEXT

Solution:

Distance travelled by car A in first two hours = 80km.
Speed = 40kmph. Ans: (40)

Directions for questions 1 and 2: Type in your answer in the input box provided below the question.

What is the difference between the average speed of car A from 1000hrs to 1230hrs and that of car B from 1230hrs to 1330hrs (In kmph)?

Solution:

Average speed (in kmph) of car A from 1000hrs to 1230hrs = $\frac{120}{2\frac{1}{2}} = \frac{120 \times 2}{5} = 48$

Average speed of car B from 1230hrs to 1330hrs = 0 (∵ car B has not travelled after 1230hrs)

∴ the required difference (in kmph) = 48 – 0 = 48 Ans: (48)

Directions for questions 3 to 5: Select the correct alternative from the given choices.

During which time period is the speed of car A the maximum?

- a) 1000hrs to 1100hrs
- b) 1100hrs to 1200hrs
- c) 1200hrs to 1300hrs
- d) 1130hrs to 1230hrs

Solution:

During 1200hrs to 1300hrs, the speed of the car A was maximum i.e. 60kmph.
Choice (C)

Directions for questions 3 to 5: Select the correct alternative from the given choices.

What was the differences in the average speed of the cars at 1230hrs?

- a) 12 kmph
- b) 15 kmph
- c) 16 kmph
- d) None of these

Solution:

At 1230hrs car A covered 120km and car B covered 160km. The average speed of A was $\frac{120 \times 2}{5}$ = 48 km/h and that of B was $\frac{160 \times 2}{5}$ = 64 km/h. The difference is 16 km/h.
Choice (C)

Directions for questions 3 to 5: Select the correct alternative from the given choices.

When car B had covered 80 km, how far ahead or behind was car A?

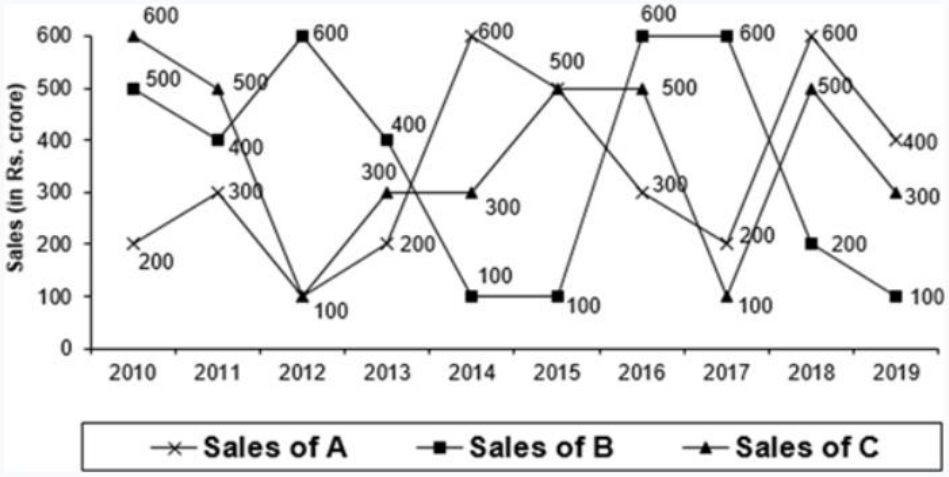
- a) 25 km ahead
- b) 15 km behind
- c) 20 km ahead
- d) 30 km behind

Solution:

Car B reached the point which is 80km away from the starting point at 1100hrs.
Car A covered 50km at 1100hrs.
∴ car A is behind car B by 30km. Choice (D)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The line graph gives the sales (in Rs.Crores) of three companies A, B and C.



Directions for questions 6 and 7: Type in your answer in the input box provided below the question.

In how many years, the sales of C was more than the sales of A but less than the sales of B?

Your Answer: Unattempted

Show Correct Answer | Hide Solution | PREVIOUS | NEXT

Solution:

This happened is 2 years 2013 and 2016. Ans: (2)

Directions for questions 6 and 7: Type in your answer in the input box provided below the question.

Find the maximum percentage increases in the sales of any of the given three companies, from 2011 to 2019 (both inclusive).

Your Answer: Unattempted

Show Correct Answer | Hide Solution | PREVIOUS | NEXT

Solution:

It can be observed that for company B, the sales increased from ₹100 crore (in 2015) to ₹600 crore (in 2016)
Required % increase = 500% Ans: (500)

Directions for questions 8 to 10: Select the correct alternative from the given choices.

In which year was the combined sales of A, B, and C, the highest?

- a) 2010
- b) 2018
- c) 2014
- d) None of these

Solution:

The total sales was maximum is year 2006. That was ₹1400 crore. Choice (D)

Directions for questions 8 to 10: Select the correct alternative from the given choices.

The average sales of which company is the maximum over the given period (2010-2019)

- a) A
- b) B
- c) C
- d) Both A and B

Show Correct Answer | Hide Solution | PREVIOUS

Solution:

The average sales is maximum when the total sales is maximum

Company	Total sales (₹crore)
A	3400
B	3600
C	3700

The average sales is the maximum for C. Choice (C)

Directions for questions 8 to 10: Select the correct alternative from the given choices.

What was the highest percentage change in the combined sales of the three companies, from one year to the next?

- a) 38.5
- b) 40.5
- c) 44.4
- d) 47.5

Solution:

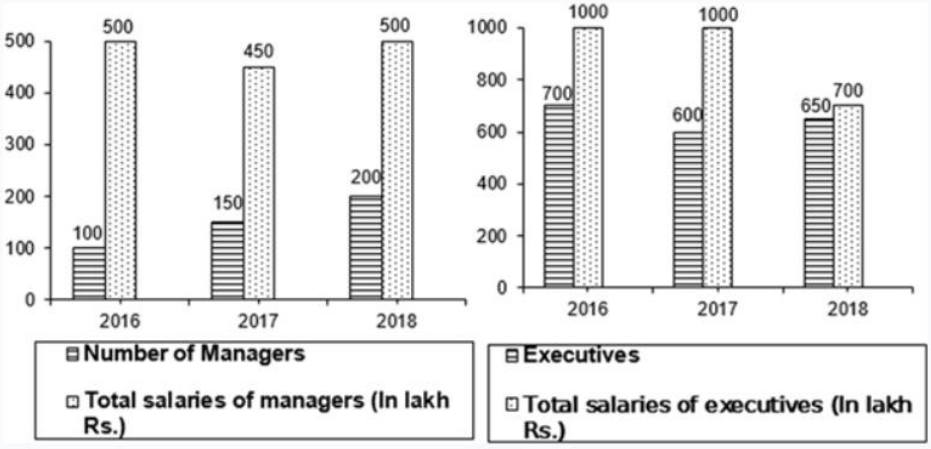
The combined sales of the three companies in the different years are as follows

2010 – 1300	2015 – 1100
2011 – 1200	2016 – 1400
2012 – 800	2017 – 900
2013 – 900	2018 – 1300
2014 – 1000	2019 – 800

The highest percentage change is in 2018 and it is $\frac{400}{900} \times 100 = 44.4$ Choice (C)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The bar graph provides information about the salaries of executives and managers in an organization across three years (Assume that the managers and the executives are the only employees in the organization).



The ratio of managers to total employees was minimum in _____.

a) 2016

b) 2017

c) 2018

d) Cannot be determined

Solution:

The ratio was minimum in 2016 i.e $\frac{1}{8}$. Choice (A)

In the year 2016 the total budget allotted to salaries was 10% of the total revenue of the company and in the year 2017, it was 12%. Find the approximate percentage increase/decrease in the revenue of the organization in 2017 when compared to that in 2016.

- a) 10.9%
- b) 14.7%
- c) 16.5%
- d) 19.4%

Solution:

Total salaries paid in 2016 = ₹1500 × 10^5
Total revenue = ₹15 × 10^7 × 10 = ₹15 × 10^8
Total salaries paid in 2017 = 1450 × 10^5

$$\text{Total revenue} = 1450 \times \frac{100}{12} \times 10^5$$

$$= \frac{145}{12} \times 10^8$$

∴ Percentage decrease

$$= \frac{15 \times 10^8 - \frac{145}{12} \times 10^8}{15 \times 10^8} \times 100 = \frac{35}{15 \times 12} \times 100 \simeq 19.44\%$$

Choice (D)

What is the percentage change in the number of employees in 2018 over that in 2017?

- a) +5%
- b) –10%
- c) $+13^1/_3\%$
- d) +10%

Solution:

Total employees in 2018 = 850 × 10^5
Total employees in 2017 = 750 × 10^5

$$\text{Percentage increase} = \frac{100}{750} \times 100 = 13\frac{1}{3}\%.$$

Choice (C)

In which year is the average salary per employee the highest?

- a) 2018
- b) 2017
- c) 2016
- d) Cannot be determined

Solution:

Clearly in the year 2017 the average salary per employee is the highest i.e., ₹1,93,300.
Choice (B)

Find the change in the average salary per employee during the given period.

- a) Rs.35,800
- b) Rs.25,000
- c) Rs.46,320
- d) Rs.19,400

Solution:

Average salary in 2016 = ₹1,87,500
Average salary in 2018 = ₹1,41,176

Change (decrease) ₹46,324

Choice (C)